Sarah Gomes Aroucha Barbosa

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Education

2021 – 2026 **Ph.D. in Physics, Federal University of Ceará (UFC), Brazil**
Machine Learning and the Search for Life: A Preparation for the Habitable Worlds Observatory.
Advisor: Prof. Daniel Brito de Freitas (UFC); Co-advisor: Dr. Raissa Estrela (JPL/Caltech).

2019 – 2021 **M.Sc. in Physics, Federal University of Ceará (UFC), Brazil**
On the metallicity-age-distance relation for F- and G-type stars in the solar neighborhood.
Advisor: Prof. Daniel Brito de Freitas.

2015 – 2019 **B.Sc. (Licentiate) in Physics, Federal Institute of Maranhão (IFMA), Brazil**
Relativistic quantum mechanics: an introduction to the Klein–Gordon field.
Advisor: Prof. Márcio da Silva Tavares.

Collaborations and Memberships

- Member, Data Processing & Analysis (DPA) Task Group, HWO AI/ML Working Group (2025 – present).
- Member, STELLAR TEAM, astrophysics group at the Federal University of Ceará.

Refereed Publications

1. **Barbosa, S. G. A.**, Estrela, R., da Silva Filho, P. C. F., Mugnai, L. V., & de Freitas, D. B. 2026, *Towards the Habitable Worlds Observatory: Retrieval of Reflection Spectra from Evolving Earth Analogues using 1D CNNs*, RAS Techniques and Instruments, HWO special issue, accepted (in proof). arXiv:2508.00076
2. **Barbosa, S. G. A.**, de Freitas, D. B., Corradi, W. J. B., da Silva Filho, P. C. F., Masuda, K., & de Medeiros, J. R. 2026, *Statistical Evidence for a Localized Projected-rotation Residual in the Solar Neighborhood*, ApJ, 1007, 66. doi:10.3847/1538-4357/ae8d27
3. **Barbosa, S. G. A.**, de Vasconcelos, B. C., da Silva, C. V., Alencar Filho, F. V., Bravo, J. P., Batista, L. F., & de Freitas, D. B. 2026, *Multiwavelength variability of Quasar 3C 273 described by nonextensive statistical mechanics*, EPL, 153, 41001. doi:10.1209/0295-5075/ae437d
4. Santiago, T. M., **Barbosa, S. G. A.**, Alves, B. W. F., Cavalcante, F. J., da Silva Filho, P. C. F., & de Freitas, D. B. 2026, *A non-extensive diagnostic of spin-orbit coupling in close-in giant exoplanets*, EPL, 155, 41001. doi:10.1209/0295-5075/ae933c
5. Santiago, T. M., **Barbosa, S. G. A.**, Cavalcante, F. J., da Silva Filho, P. C. F., & de Freitas, D. B. 2025, *A new non-extensive proxy for the evolution of stellar rotation period in planetary systems*, EPL, 151, 29001. doi:10.1209/0295-5075/adecbf
6. Lima Filho, F. J. S., Ferreira, V. M. B., da Silva Filho, P. C. F., Gomes, F. O. S., Alves, B. W. F., **Barbosa, S. G. A.**, Santiago, T. M., & de Freitas, D. B. 2022, *Multifractal characterization as a function of timescale in the light curves with planetary signal observed by the Kepler mission*, Brazilian Journal of Development, 8, 67992.

Preprints and Community Papers

1. Dressing, C. D., et al. (including **Barbosa, S. G. A.**) 2026, *Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory*, HWO community paper, to be submitted. arXiv:2608.11294

Conference Proceedings and Books

1. **Barbosa, S. G. A.**, Estrela, R. L. F., da Silva Filho, P. C. F., & de Freitas, D. B. 2026, *Convolutional Neural Networks for Characterizing Reflection Spectra of Evolving Earth-Like Planets*, in HWO25 Proceedings Part II: Mission Framework, Technology, and Broader Contributions, ASP Conf. Ser., 543, 131.
2. Tavares, M. S., & **Barbosa, S. G. A.** 2020, *Mecânica Quântica Relativística: Uma Introdução ao Campo de Klein-Gordon* (book, in Portuguese), Novas Edições Acadêmicas, ISBN 978-6200579942.

Talks and Presentations

- 2026: *Machine Learning and the Search for Life: A Preparation for the Habitable Worlds Observatory*. From Stars to Life: Stellar Impacts on Planetary Atmospheres, Habitability, and Biosignatures Detection, Natal, Brazil.
- 2025: *Convolutional Neural Networks for Characterizing Reflection Spectra of Evolving Earth-Like Planets*, Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology (STScI), Johns Hopkins Bloomberg Center, Washington, DC, USA.
- 2024: *Deep learning for analyzing reflected spectra from Earth analogs throughout their geological evolution around FGKM stars*, oral communication, XLVII Annual Meeting of the Brazilian Astronomical Society (SAB), Águas de Lindóia, Brazil.

Observational Experience

- 2025 – present: Team member (PI: da Silva Filho, P. C. F.), SPARC4 program at the Pico dos Dias Observatory (OPD/LNA, Brazil) to recover the orbital period of the long-period transit candidate TOI-706 (HD 32226), using transit models built from TESS sectors 4-6, 31, and 32 to predict the observing windows.

Supervision and Mentoring

- 2026 – present: Eduardo Assunção (undergraduate research), *Homogeneous C, N, and O abundances for Habitable Worlds Observatory target stars*, UFC.
- 2026 – present: Josué Araújo (undergraduate research), *Spectral window dependence in stellar parameter inference of M dwarfs*, UFC.
- 2025: Co-supervisor (with D. B. de Freitas) of Maggie Henrique Araújo Soares, undergraduate final project, *Classification of rocky exoplanets using artificial intelligence: a research-initiation approach in high school*, UFC.

Professional Service

- 2026 – present: Referee for *Astrophysics and Space Science*.
- 2023: Examining committee, undergraduate thesis on students' perception of artificial intelligence in physics teaching, UFC.

Languages

Portuguese (native); English (advanced in reading and writing; intermediate in speaking); Spanish (intermediate).